Linear and Binary Search Algorithms

Search Algorithms

Search algorithms are used to find items of data. There are some standard algorithms that have been developed and each have their own merits. Over the next few pages we will take a look at two common search algorithms:

- Linear Search Algorithm
- Binary Search Algorithm

Linear Search

Linear searching simply involves searching through a set of data, one item after the other, until the item we are looking for is found. Let's take a look at an example of using a linear search algorithm to find an item of data in an array.

Imagine we are searching for the letter 'D'.

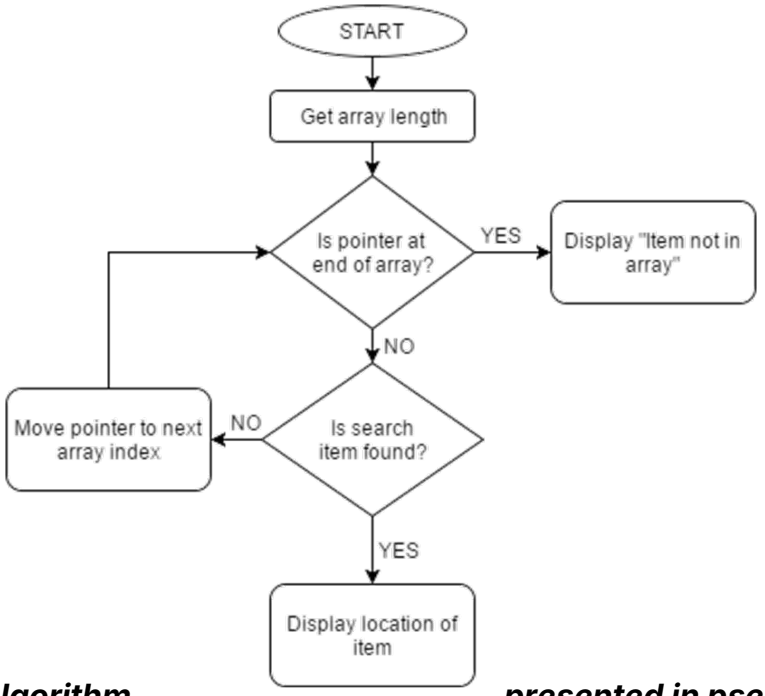
INDEX	0	1	2	3	4	5	6
Item	A	F	B	E	D	G	C

A linear search algorithm will be set up to cycle through each index of the array.

IF it finds a 'D', it knows to stop and display the location.

ELSE it will add 1 to the index (move to the next item) and look there.

The Linear Search Algorithm presented as a flowchart:



The Linear Search Algorithm presented in pseudocode:

```
position = 0
found = False
WHILE (position < length(array)) AND (found == False):
        IF search_item == array[counter]
                DISPLAY "Item at position" counter+1
                found = True
        ELSE:
                position = position + 1
```

The Binary Search Algorithm (a *DIVIDE AND CONQUER* algorithm)

Binary searching is only possible for sets of data that are in ORDER – THEY HAVE TO BE ORDERED!!

It involves starting the search at the middle to work out if the search item is positioned lower than the middle or higher than the middle.

If it is positioned lower than the middle, the search is narrowed to the first half of the data set.

If it is positioned higher than the middle, the search is narrowed to the second half of the data set.

Once the data set is halved, the search then continues as it did before, in the middle.

By continually dividing the data set until the item is found, the time taken to search for an item can be much faster than using a linear search.

An Example

Step 1

Lets say that we are searching for the letter 'G.

We would begin by looking at the middle item.

To do this we would identify the lower bound and the upper bound of the data set.

INDEX	0	1	2	3	4	5	6	7	8	9	10
Item	A	B	C	D	E	F	G	H	I	J	K
	LB										UB

Then we would add the 'lower bound' to the 'upper bound' and divide by 2, in order to find the position of the middle item (mid-point): (10-0)/2 = Midpoint = 5

The item at position 5 is the letter "F".

"F" is lower than the letter "G".

We now narrow our data set by focusing on the second half of the set.

Step 2

Computer Science UK Membership Site Licence: Do not share outside your centre.

The lower-bound moves to the position one up from the mid-point...

INDEX	0	1	2	3	4	5	6	7	8	9	10
Item	A	B	C	D	E	F	G	H	I	J	K
							LB				UB

...and we repeat the same process again.

We add the 'lower bound' to the 'upper bound' and divide by 2, in order to find the position of the middle item (mid-point): (10+6)/2 = Midpoint = 8

The item at position 8 is the letter "I"..."I" is higher than the letter "G".

We now narrow our data set by focusing on the first half of this new, smaller set.

Step 3

The lower-bound moves to the position one down from the mid-point...

INDEX	0	1	2	3	4	5	6	7	8	9	10
Item	A	B	C	D	E	F	G	H	I	J	K
							LB	UB			

...and we repeat the same process again.

We add the 'lower bound' to the 'upper bound' and divide by 2, in order to find the position of the middle item (mid-point): (7+6)/2 = Midpoint = 6.5

ISSUE! WE DON'T HAVE AN INDEX 6.5!

OK, what we do is round the answer up to 7 and use this as our midpoint.

The item at position 7 is the letter "H"..."H" is higher than the letter "G".

We now narrow our data set by focusing on the first half of this new, smaller set.

Step 4

The lower-bound moves to the position one down from the mid-point...

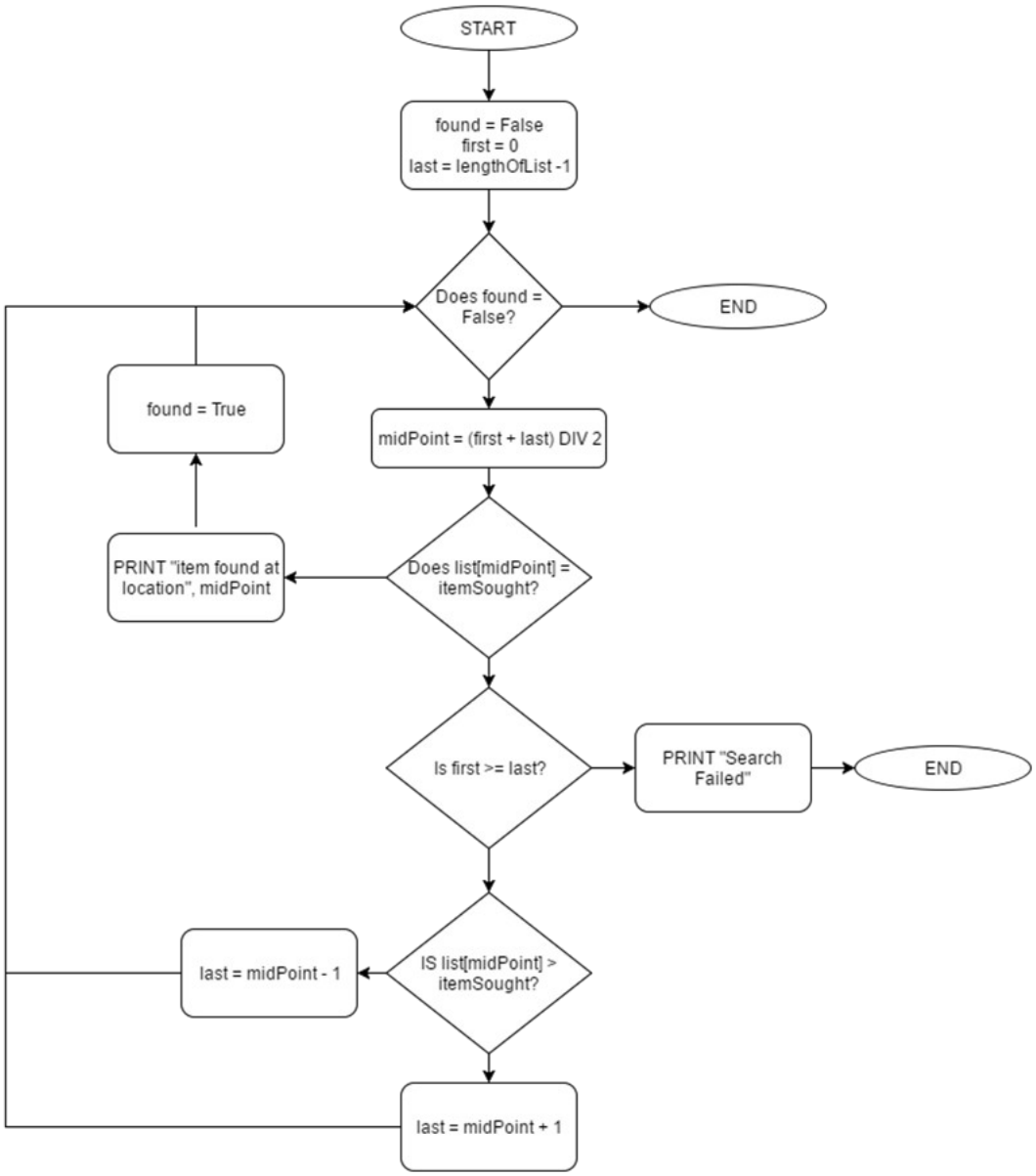
INDEX	0	1	2	3	4	5	6	7	8	9	10
Item	A	B	C	D	E	F	G	H	I	J	K
							LB				
							UB				

...and we repeat the same process again.

We add the 'lower bound' to the 'upper bound' and divide by 2, in order to find the position of the middle item (mid-point): (6+6)/2 = Midpoint = 6

The item at position 6 is the letter "G". We have now found them item we were searching for!!

The Binary Search Algorithm presented as a flowchart:



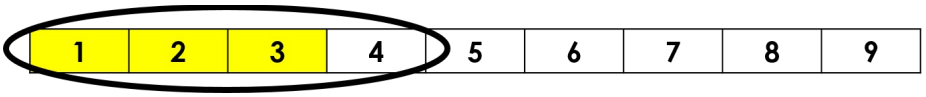
The Binary Search Algorithm presented in pseudocode:

```
found = False
first = 0
last = lengthOfList – 1
WHILE found = False:
        midPoint = (first + last) DIV 2
        IF list[midpoint] == itemSought:
                PRINT "item found at location", midpoint
                found = True
                BREAK
        ELSE:
                IF first >= last:
                        PRINT "Failed"
                        BREAK
                ELSE:
                        IF list[midpoint] > itemSought:
                                last = midpoint - 1
                        ELSE:
                                first = midpoint + 1
```

Questions *(The question zone you choose must either match your target grade or be higher!)*

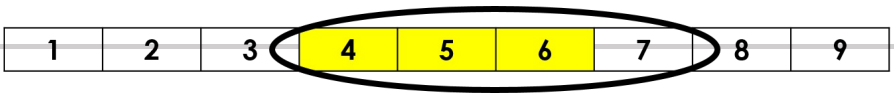
Computer Science UK Membership Site Licence: Do not share outside your centre.

Question Zone 1-3



1. Describe how a linear/serial search works. [2]
2. How must a dataset be prepared before a binary search can effectively search through the data? [2]
3. Describe how a binary search works. [5]

Question Zone 4-6



1. Describe how a binary search works. [5]
2. Perform a binary search in order to find the number 10, in the following dataset, clearly showing each step you take.

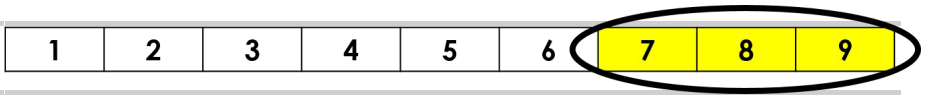
   [3, 7, 10, 11, 13, 15]

   [3]
3. Perform a binary search in order to find the number 15, in the following dataset, clearly showing each step you take. Prepare the dataset before you begin.

   [10, 14, 3, 11, 7, 13]

   [3]

Questions Zone 7-9



1. Perform a binary search in order to find the number 15, in the following dataset, clearly showing each step you take. Prepare the dataset before you begin.

   [10, 14, 3, 11, 7, 13]

   [3]
2. Show the steps that both a linear search and a binary search would take to find the number 20 in the following dataset:

   [2, 4, 7, 12, 15, 18, 20]

   [5]
3. Which search algorithm is more efficient, a linear search or a binary search? Explain your answer fully. [5]

Computer Science UK Membership Site Licence: Do not share outside your centre.

Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

Linear/Serial Algorithm: *A method of searching for an item in a dataset by systematically checking each item in order until the searched for item is found.*

Binary Search Algorithm: *A method of searching for an item in a **sorted** dataset by repeatedly comparing the middle item with the item being searched for and discarding the half of the dataset which doesn't contain the item, until the item is found.*

Algorithm*: A set of steps/instructions needed to complete a software task.*

Computer Science UK Membership Site Licence: Do not share outside your centre.

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**!

Computer Science UK Membership Site Licence: Do not share outside your centre.

Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy, as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

_____

_____

_____

_____

Computer Science UK Membership Site Licence: Do not share outside your centre.